Network Security Laboratory

Experiment 5: ARP Spoofing and Detection

1. Objective

Perform a simple ARP spoofing test inside an isolated lab network and detect the change using the victim ARP table and Wireshark.

2. Required Programs

- Kali Linux on the attacker laptop
- Wireshark on the victim laptop
- dsniff package (arpspoof tool)

3. Topology

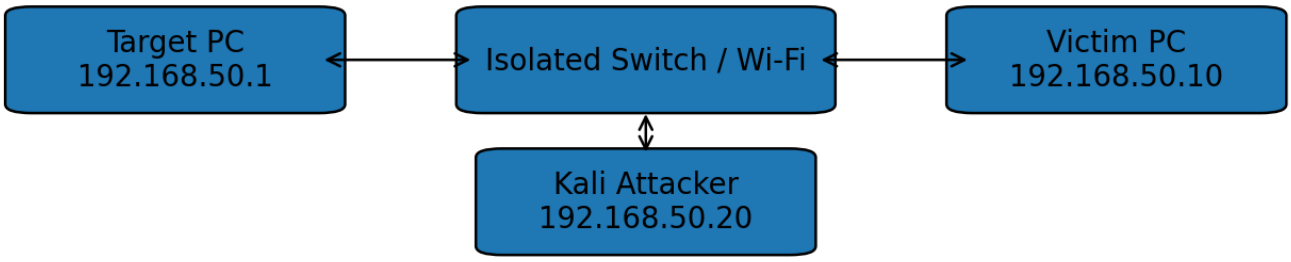


Figure 1. Isolated lab topology

Device	IP Address
Target PC	192.168.50.1
Victim PC	192.168.50.10
Kali Attacker	192.168.50.20

Subnet Mask: 255.255.255.0

4. Before the Attack

On the victim laptop:

```
ping 192.168.50.1
arp -a
```

Record the MAC address associated with 192.168.50.1. The ping should work normally.

5. Start Wireshark

- Open Wireshark on the victim laptop.
- Select the active network interface.
- Start packet capture.
- Use the display filter below:

```
arp
```

6. Start ARP Spoofing

On Kali Linux, use the correct interface name (for example eth0):

```
sudo apt install dsniff -y
sudo arpspoof -i eth0 -t 192.168.50.10 192.168.50.1
```

This repeatedly sends forged ARP replies to the victim, claiming that 192.168.50.1 is associated with the Kali laptop MAC address.

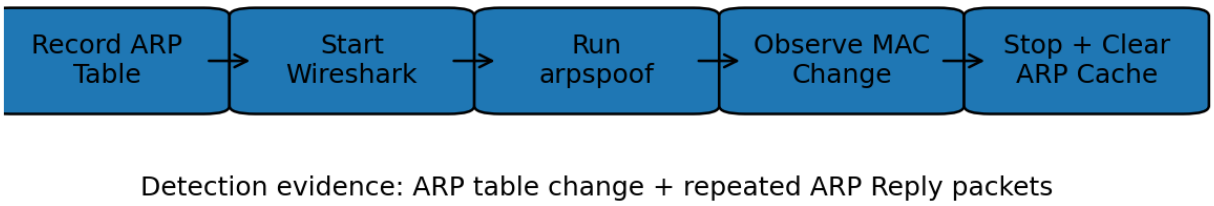


Figure 2. Simplified test and detection flow

7. Detect the Attack

On the victim laptop:

```
arp -a
ping 192.168.50.1
```

Expected evidence:

- The MAC address for 192.168.50.1 changes to the Kali MAC address.
- The ping may fail or behave abnormally.
- Wireshark shows repeated ARP Reply packets similar to:

```
192.168.50.1 is at Kali_MAC_Address
```

8. Stop and Remove the Test

On Kali Linux:

```
Ctrl + C
```

On the victim laptop, open Command Prompt as Administrator:

```
arp -d *
ping 192.168.50.1
arp -a
```

The correct MAC address should appear again and the ping should return to normal.

9. Student Task

- Record the ARP table before the test.
- Start Wireshark and apply the arp filter.
- Run arpspoof from Kali.
- Capture the changed MAC address.
- Capture repeated ARP Reply packets in Wireshark.
- Stop the test and clear the ARP cache.
- Verify the correct MAC address is restored.

10. Required Screenshots

- ARP table before the attack.
- arpspoof command on Kali.
- ARP packets in Wireshark.
- Changed MAC address during the attack.
- Restored ARP table after stopping the attack.

11. Quick Questions

What is ARP spoofing?

Sending forged ARP replies so an IP address is associated with the wrong MAC address.

How was the attack detected?

By the changed MAC address in the ARP table and repeated ARP replies in Wireshark.

How was the ARP cache cleared?

Using arp -d *.

Submission

Save the report as: Week5_ARP_StudentName.pdf